```
vboxuser@Kartik:~$ nano exp7a.c
vboxuser@Kartik:~$ gcc exp7a.c -o exp7a
vboxuser@Kartik:~$ chmod +x exp7a
vboxuser@Kartik:~$ ./exp7a
```

```
  GNU nano 8.7.1
#include<stdio.h>

int main() {
    long long counter = 0;

    while (1) {
        counter++;
        if(counter % 100000000 == 0) {
            printf("CPU task running: %lld\n", counter);
        }
    }
        return 0;
}
```

```
vboxuser@Kartik:~$ pgrep -a exp7a
7182 ./exp7a
vboxuser@Kartik:~$ top -p 7182
```

```
top - 14:58:52 up 19 min,  1 user,  load average: 0.67, 0.36, 0.24
Tasks:   1 total,   1 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s): 10.9 us,  3.1 sy,  0.0 ni, 84.4 id,  0.0 wa,  0.0 hi,  1.6 si,  0.0 st
MiB Mem :   4407.5 total,    676.1 free,   1439.3 used,   2571.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   2968.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   7182 vboxuser  20   0    2768   1748   1648 R 100.0   0.0   0:50.88 exp7a
```

```
top - 06:42:56 up 23 min,  1 user,  load average: 1.42, 1.11, 0.72
Tasks:   1 total,   1 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s):  4.4 us,  0.5 sy,  0.0 ni, 95.0 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :   4407.5 total,   1546.2 free,   1415.9 used,   1721.0 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   2991.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   5661 vboxuser  20   0    2768   1744   1648 R  99.7   0.0   0:19.92 exp7a
```

```
vboxuser@Kartik:~$ nano exp7b.c
vboxuser@Kartik:~$ gcc exp7b.c -o exp7b
vboxuser@Kartik:~$ chmod +x exp7b
vboxuser@Kartik:~$ ./exp7b
```

```
  GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main() {
    FILE *fp;
    printf("Starting I/O task...\n");
    while (1) {
        fp = fopen("dummy.txt", "w");
        if (fp != NULL) {
            fprintf(fp, "Writing disk data\n");
            fclose(fp);
            printf("File operation done\n");
          }
          sleep(1);
          }
          return 0;
}
```

```
vboxuser@Kartik:~$ pgrep -a exp7b
7424 ./exp7b
vboxuser@Kartik:~$ top -p 7424
```